



CSE-476: Introduction to Natural Language Processing  
Quiz 2

## Multiple Choice Questions (Single correct)

*Instructions: Select the correct option with a short explanation.*

- Q1. What is the main difference between generative and discriminative classifiers?
- (A) Generative models model  $P(x, y)$ , while discriminative models model  $P(y | x)$
  - (B) Discriminative models learn the data distribution and use that knowledge for classification
  - (C) Discriminative models require no training data
  - (D) Generative models are less explainable
- Q2. Which technique is used to mitigate overfitting by penalizing large weights?
- (A) Laplace smoothing
  - (B) Regularization
  - (C) TF-IDF
  - (D) Softmax normalization
- Q3. What is the limitation of the perceptron algorithm?
- (A) It cannot use vector representations
  - (B) It only works for linearly separable data
  - (C) It requires probabilistic outputs
  - (D) It cannot be trained with gradient descent
- Q4. Why is Laplace smoothing important?
- (A) It improves cosine similarity
  - (B) It reduces dimensionality
  - (C) It avoids zero probabilities for unseen words
  - (D) It speeds up gradient descent
- Q5. What happens in stochastic gradient descent (SGD) compared to gradient descent (GD)?
- (A) SGD updates weights using the entire dataset every step

- (B) SGD updates weights using one or a subset of training examples
  - (C) GD converges faster because updates are noisier
  - (D) SGD cannot be used with neural networks
- Q6. What is the core intuition behind word embeddings?
- (A) Words are represented by their dictionary definitions
  - (B) Words that appear in similar contexts have similar meanings
  - (C) Words are clustered by part of speech
  - (D) Words are ranked by frequency
- Q7. What is the primary purpose of the Softmax function?
- (A) To penalize large model weights
  - (B) To reduce feature dimensionality
  - (C) To select the class with the largest raw score without normalization
  - (D) To convert raw class scores into a valid probability distribution
- Q8. In the perceptron algorithm, when are the weights updated?
- (A) After every training example
  - (B) Only when the prediction is correct
  - (C) Only when the prediction is incorrect
  - (D) Only after one full pass over the data
- Q9. What is the main goal of TF-IDF weighting?
- (A) Increase the weight of very common words
  - (B) Represent word order
  - (C) Convert sparse vectors into dense embeddings
  - (D) Highlight words that are frequent in a document but rare across documents
- Q10. Why is cosine similarity preferred over dot product when comparing word vectors?
- (A) It ignores vector length while calculating similarity
  - (B) It increases the influence of high-frequency words
  - (C) It converts similarity scores into probabilities
  - (D) It reduces the dimensionality of word embeddings

*End of Paper*